

Matrix Normalization

What is Matrix Normalization?

Matrix normalization is when we modify the values of the matrix to be between in a specific range like (0→1) to connect the rows with each other or even the columns with each other by a specific connection

What are the types of Normalization?

1. Min-Max Normalization
2. Z-score Normalization
3. L2 Normalization

1-Min-Max Normalization

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

1. We use it when we want to set the values to range (usually 0 to 1) because we have different sets of information so the impact will be fair.
 2. Neural Networks to decrease the time of training
 3. Image processing to change the pixels from (0→255) to (0→1) by dividing by 255
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2-Z-score Normalization

$$Z = \frac{X - \mu}{\sigma}$$

Make data have:

- Mean = 0
- Standard deviation = 1

1. We use it in Neural Networks because it learns faster when the data are around zero
2. With Gradient Descent because if the values are big the gradient will be slow and wavering so after the standardization the convergence will be faster.
3. With Linear Regression and Logistic Regression to make the weights comparable

- ❖ WE USE IT WHEN THE DATA ISN'T BETWEEN FIXED RANGE
 - ❖ WHEN IT CONTAINS OUTLIERS
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3-L2 Normalization

$$x_{norm} = \frac{x}{\|x\|_2}$$

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + x_3^2 + \dots + x_m^2}$$

Scaling a vector so that its length becomes 1

- ❖ It keeps the direction the same, but changes the size
1. We use it in Cosine Similarity in (NLP, Searching)
 2. Computer Vision
 3. Deep Learning
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➤ Summary

Min-Max → Range

Z-Score → Mean and deviation

L2 → Length of Vector

○ Sources and references:

1. <https://youtu.be/sxEqtjLC0aM?si=l2blAGmCqpFZiwNz>
2. <https://youtu.be/7fn03DIW3Ak?si=AnATOTN-7wx2E7Em>
3. ChatGPT